Day 129 | Unit 09 | Teacher plan

### Display, Describe, and Summarize Data

# Choose Appropriate Statistical Measures

Standards: 6.SP.B.5d

Learning target: I can choose a measure of center and explain why.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

## 0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

## 5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Use the shape and context. Mean with mean absolute deviation can describe fairly symmetric data. Median with interquartile range often describes skewed data or data with an unusual value more fairly.

## 17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

## 42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

## Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

## Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references:

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## 01 / Learn

Use the shape and context. Mean with mean absolute deviation can describe fairly symmetric data. Median with interquartile range often describes skewed data or data with an unusual value more fairly.

## Worked example 1

Reading times in minutes are 5, 6, 6, 7, 7, 8. Choose a center and matching variability measure. Calculate both and justify your choice.

1. Use the shape and context.

2. Mean with mean absolute deviation can describe fairly symmetric data.

3. Mean = 13/2, median = 13/2, IQR = 7 - 6 = 1, MAD = 5/6.

4. Accept another justified choice tied to shape and purpose.

5. State the result in the form requested: Mean 13/2 min and MAD 5/6 min; roughly symmetric data.

Answer: Mean 13/2 min and MAD 5/6 min; roughly symmetric data

## Worked example 2

Six students recorded reading times (minutes): 3, 9, 9, 15, 15, 21. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. Use the shape and context.

2. Order the observations.

3. Middle positions give median 12 and half medians give IQR 6.

4. Mean is 72/6

5. averaging distances gives MAD 5.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 12, IQR 6; mean 12, MAD 5. Roughly balanced; mean/MAD or median/IQR with justification.

# Day 129 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

### 01 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Choose a center and matching variability measure. Calculate both and justify your choice.

### 02 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 8, 9. Choose a center and matching variability measure. Calculate both and justify your choice.

### 03 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Choose a center and matching variability measure. Calculate both and justify your choice.

### 04 / Constructed response

Reading times in minutes are 2, 8, 8, 14, 14, 20. Choose a center and matching variability measure. Calculate both and justify your choice.

### 05 / Constructed response

Reading times in minutes are 6, 10, 10, 14, 14, 18. Choose a center and matching variability measure. Calculate both and justify your choice.

## 06 / Constructed response

Reading times in minutes are 5, 9, 9, 13, 13, 17. Choose a center and matching variability measure. Calculate both and justify your choice.

## 07 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 6, 15. Choose a center and matching variability measure. Calculate both and justify your choice.

## 08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 4, 8, 8, 12, 12, 16. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

## 09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 19. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

## 10 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 10, 19. Choose a center and matching variability measure. Calculate both and justify your choice.

Day 129 | Unit 09 | Answer key

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Learning target: I can choose a measure of center and explain why.

### 1. Mean 11/2 min and MAD 5/6 min; roughly symmetric data

Mean = 11/2, median = 11/2, IQR = 6 - 5 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

### 2. Mean 15/2 min and MAD 5/6 min; roughly symmetric data

Mean = 15/2, median = 15/2, IQR = 8 - 7 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

### 3. Mean 9/2 min and MAD 5/6 min; roughly symmetric data

Mean = 9/2, median = 9/2, IQR = 5 - 4 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

### 4. Mean 11 min and MAD 5 min; roughly symmetric data

Mean = 11, median = 11, IQR = 14 - 8 = 6, MAD = 5. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

### 5. Mean 12 min and MAD 10/3 min; roughly symmetric data

Mean = 12, median = 12, IQR = 14 - 10 = 4, MAD = 10/3. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

### 6. Mean 11 min and MAD 10/3 min; roughly symmetric data

Mean = 11, median = 11, IQR = 13 - 9 = 4, MAD = 10/3. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

### 7. Median 9/2 min and IQR 2 min; high value makes these resistant measures useful

Mean = 37/6, median = 9/2, IQR = 6 - 4 = 2, MAD = 53/18. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

## 8. 6 observations; minutes; median 10, IQR 4; mean 10, MAD 10/3. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 10 and half medians give IQR 4. Mean is 60/6; averaging distances gives MAD 10/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

## 9. 6 observations; minutes; median 13/2, IQR 3; mean 47/6, MAD 23/6. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 13/2 and half medians give IQR 3. Mean is 47/6; averaging distances gives MAD 23/6. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

## 10. Median 17/2 min and IQR 2 min; high value makes these resistant measures useful

Mean = 61/6, median = 17/2, IQR = 10 - 8 = 2, MAD = 53/18. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response